

SYDNEY GIRLS HIGH SCHOOL



YEAR 11 Mathematics

Assessment Task 3

September 2015

Time allowed: 60 minutes
Plus 5 minutes reading time
Chapters : 1- 12

Instructions:

- There are Four (4) questions. Questions **are** of equal value.
- Attempt all questions.
- Show all necessary working. Marks may be deducted for badly arranged work.
- Start each question on a new page. Write on one side of the paper only.

Student Name:-----

Teacher:-----

QUESTION ONE (15 marks)

a) Given $f(x) = x^2 - 3x$, find $f(2)$. (1)

b) Simplify $1 - \frac{4-a}{3}$ as single fraction (1)

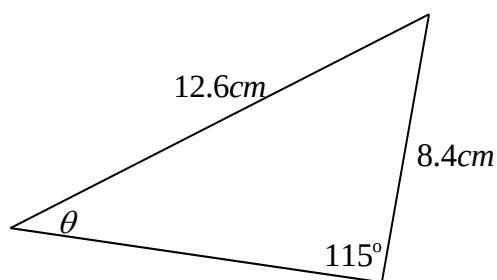
c) Factorise completely $8 - 18t^2$ (1)

d) If $(3\sqrt{12} - 2\sqrt{5})^2 = a + b\sqrt{15}$, find a and b . (2)

e) Solve $|2x - 3| \leq 11$. (2)

f) Find $\lim_{x \rightarrow 4} \frac{x-4}{x^2-16}$. (1)

g) Find the value of θ , to the nearest degree. (2)



h) Simplify $\frac{4}{x-3} - \frac{2x+1}{x^2-5x+6}$ (3)

i) Evaluate $(1 + \tan 60^\circ)^2$, in exact form. (2)

QUESTION TWO (15 marks)

a) Sketch the following, showing all the main features

i) $y = -x^2 - 5x$ (2)

ii) $y = |x + 4|$ (1)

iii) $y = \sqrt{9 - x^2}$ (1)

b) Solve $\sqrt{5} \sin \theta = -1$, for $0^\circ \leq \theta \leq 360^\circ$. (answer to the nearest degree) (2)

c) Differentiate the following

i) $y = 6x^3 + 2x^{-4} - 10$ (1)

ii) $y = (7 - 3x^2)^7$ (1)

iii) $y = x^2 \sqrt{x}$ (1)

d) Graph the region bounded by the following $2x + y < -1$ and $y \geq x^2 - 4$ (2)

e) Show that $f(x) = \frac{2x}{x^2 + 1}$ is an odd function. (2)

f) Find the domain of $y = \sqrt{4 - x} + \sqrt{-2x + 7}$ (2)

QUESTION THREE (15 marks)

a) On your answer sheet plot the points $P(0,10)$, $Q(-5,0)$ and $R(2,-2)$ which are the vertices of $\triangle PQR$. The origin is $O(0,0)$.

i) Find the exact length of PQ . (1)

ii) Show that PQ has equation $2x - y + 10 = 0$. (2)

iii) Find the perpendicular distance from R to PQ . (2)

iv) Find the coordinates of S such that $PQRS$ is a parallelogram. (1)

v) Find the area of $PQRS$. (1)

vi) Find, correct to the nearest minute, the size of $\angle PQO$. (2)

b) Differentiate

i) $y = \frac{3x^2 - 4}{8 - 5x}$ (2)

ii) $y = (10 - 3x)\sqrt{1 - 4x^2}$ (2)

c) Show that $\frac{\sin(270^\circ - \theta)}{\sin \theta} = -\cot \theta$. (2)

QUESTION FOUR (15 marks)

a) Find the equation of the tangent to the curve $y = x^2 - 8x + 20$ at $x = 2$. (3)

b) Prove that $\frac{\sin \theta}{1 - \sin \theta} - \frac{\sin \theta}{1 + \sin \theta} = 2 \tan^2 \theta$. (2)

c) Bella is running from Jacob at a speed of 300 km/ hr and after 10 minutes she is at a bearing of $130^\circ T$ from Jacob. Edward is waiting for Bella 37 km due north-east of Jacob.

- Draw a diagram showing the above information. (1)
- Find the distance between Bella and Edward. (correct to 1d.p) (2)
- Calculate the bearing of Edward from Bella. (to the nearest degree) (2)

d)

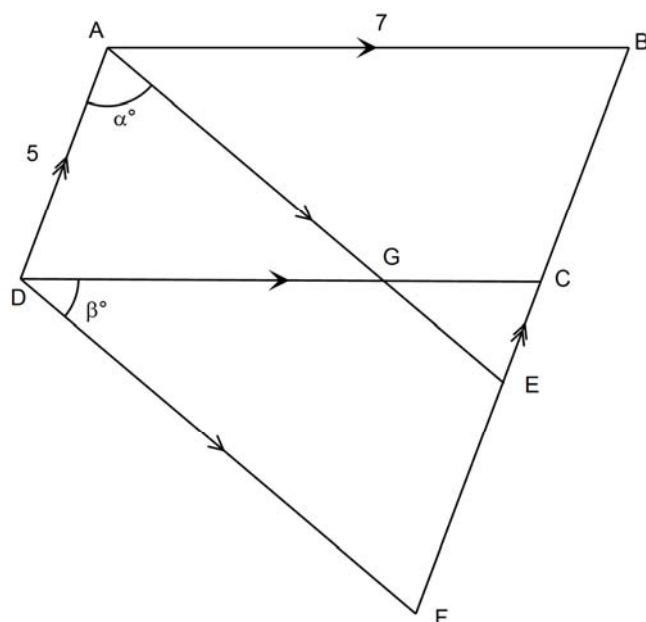


Figure not to scale

In the diagram above, ABCD and AEFD are parallelograms. AE bisects $\angle DAB$ and DC bisects $\angle FDA$. Let $\angle DAG = \alpha$ and $\angle FDG = \beta$.

- Copy the diagram on your answer sheet
- Show that $\triangle AGD$ is equilateral, giving reasons. (2)
- If $AD = 5$ units and $AB = 7$ units, find the area of the trapezium $ABFD$. (3)
(Giving your answer in simplest exact form)

THE END

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2015 Unit Yr11 Yearly.

Q, a) $f(x) = x^2 - 3x$

$$f(2) = 2^2 - 3 \times 2$$

$$f(2) = -2$$

b) $1 - \frac{4-a}{3}$

Common mistake due to -ve sign

$$= \frac{3}{3} - \frac{4-a}{3}$$

$$= \frac{3 - (4-a)}{3}$$

$$= \frac{3-4+a}{3}$$

$$= \frac{a-1}{3}$$

c) $8 - 18t^2$

$$= 2(4 - 9t^2)$$

$$= 2(2+3t)(2-3t)$$

d) $(3\sqrt{12} - 2\sqrt{5})^2$

$$= 9 \times 12 - 12\sqrt{60} + 20$$

$$= 108 + 20 - 12\sqrt{60}$$

$$= 128 - 12\sqrt{60}$$

$$= 128 - 24\sqrt{15}$$

$$\therefore a = 128 \quad b = -24$$

$$e) |2x-3| \leq 11$$

$$-11 \leq 2x-3 \leq 11$$

$$-8 \leq 2x \leq 14$$

$$-4 \leq x \leq 7$$

$$f) \lim_{x \rightarrow 4} \frac{(x-4)}{(x-4)(x+4)}$$

$$= \lim_{x \rightarrow 4} \frac{1}{x+4}$$

$$= \frac{1}{4+4} = \frac{1}{8}$$

Some students failed to factorise before substitution.

$$g) \frac{\sin \theta}{8.4} = \frac{\sin 115^\circ}{12.6}$$

$$\sin \theta = \frac{8.4 \sin 115^\circ}{12.6}$$

$$\theta = 37^\circ$$

Students used sine rule, sine ratio which incorrect.

$$h) \frac{4}{x-3} - \frac{2x+1}{(x-3)(x-2)}$$

$$= \frac{4(x-2)}{(x-3)(x-2)} - \frac{2x+1}{(x-3)(x-2)}$$

$$= \frac{4x-8-2x-1}{(x-3)(x-2)}$$

$$= \frac{2x-9}{(x-3)(x-2)}$$

Common mistake due to -ve sign

$$\begin{aligned} \text{i)} \quad & (1 + \tan 60^\circ)^2 \\ &= (1 + \sqrt{3})^2 \\ &= 1 + 2\sqrt{3} + 3 \\ &= 4 + 2\sqrt{3} \end{aligned}$$

Mathematics 2015 yearly

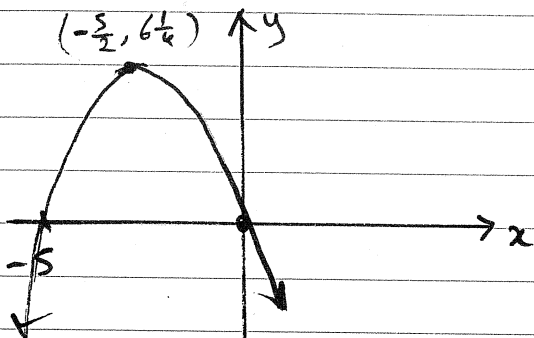
2a)

$$i) y = -x(x+5)$$

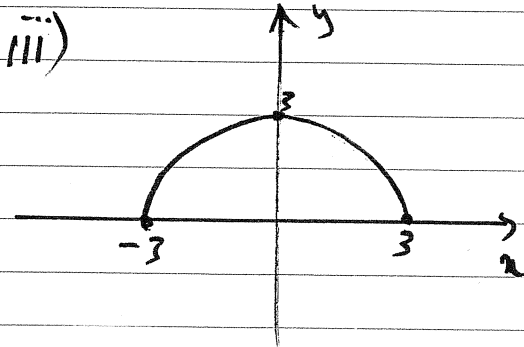
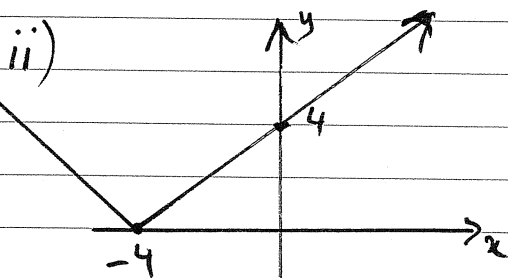
$$x=0 \text{ or } -5$$

$$\text{at } x = -\frac{5}{2}, y = 6\frac{1}{4}$$

$$(-\frac{5}{2}, 6\frac{1}{4})$$



* many students didn't show the vertex



b) $\sin \theta = -\frac{1}{\sqrt{5}}$

$$\theta = 206^\circ, 333^\circ$$

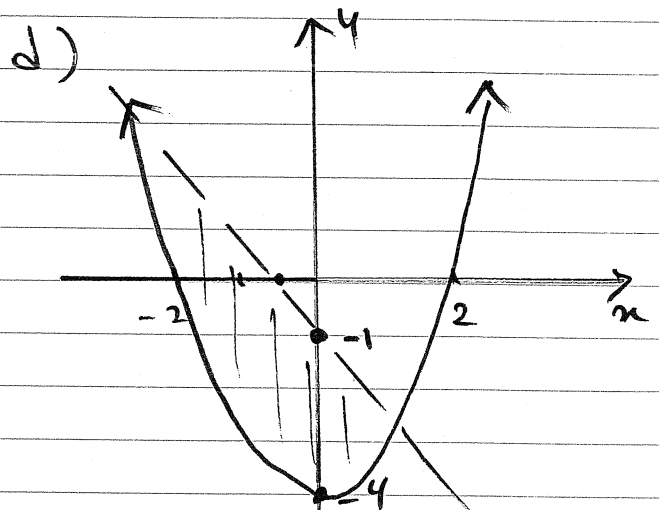
* some students couldn't get the 2nd answer

c) i) $y' = 18x^2 - 8x^{-5}$

ii) $y' = 7(7-3x^2)^6 \cdot -6x$
 $= -42x(7-3x^2)^6$

iii) $y = x^{\frac{5}{2}}$
 $y' = \frac{5}{2} x^{\frac{3}{2}}$

* some students were careless in working with indices



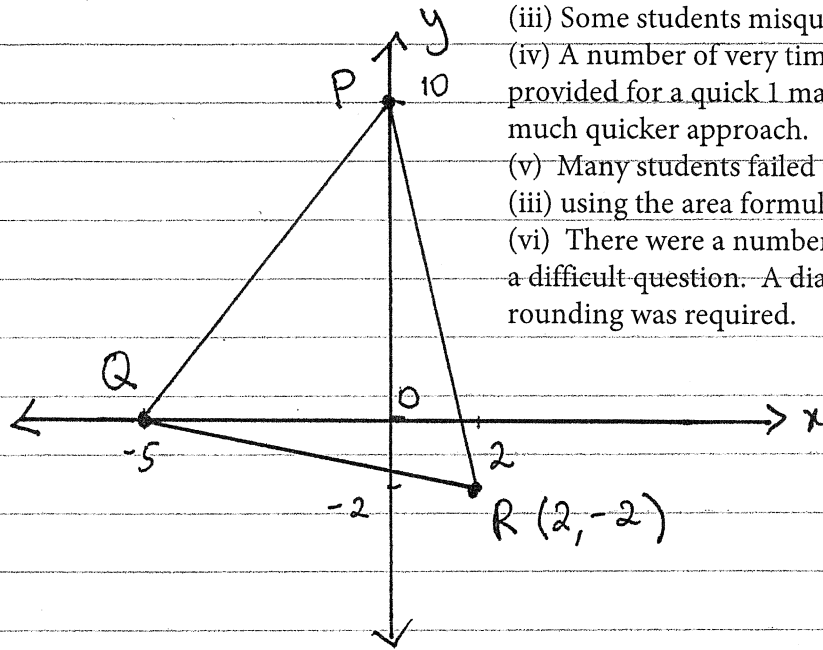
* Some student didn't use dotted line and didn't have the correct closed region. Some didn't have any numbers on the number plane. Very Poorly done.

e) $f(-x) = \frac{2(-x)}{(-x)^2+1} = \frac{-2x}{x^2+1}$
 $-f(x) = \frac{-2x}{x^2+1} \quad f(x) = \frac{2x}{x^2+1} \therefore \text{odd}$

f) $4-x \geq 0 \therefore x \leq 4$
 $-2x+7 \geq 0 \therefore x \leq 3\frac{1}{2}$
 $\therefore D: x \leq 3\frac{1}{2}$ very poorly done.

QUESTION THREE

(a)



$$(i) PQ = \sqrt{5^2 + 10^2}$$

$$\therefore PQ = \sqrt{125} \text{ or } 5\sqrt{5} \text{ units}$$

$$(ii) \text{ From diagram } m_{PQ} = \frac{\text{rise}}{\text{run}} = \frac{10}{5} = 2$$

y-int. of PQ is 10

$$\therefore PQ \text{ has equation } y = 2x + 10$$

$$\text{i.e. } 2x - y + 10 = 0$$

$$(iii) d_{\perp} = \frac{|2(2) - (-2) + 10|}{\sqrt{2^2 + (-1)^2}} = \frac{16}{\sqrt{5}} \text{ units}$$

$$(iv) S = (7, 8)$$

$$(v) A = bh = 5\sqrt{5} \times \frac{16}{\sqrt{5}} = 80 \text{ units}^2$$

$$(vi) \tan \theta = m \quad \therefore \tan \angle PQO = 2$$
$$\angle PQO \doteq 63^{\circ} 26'$$

Comments for part (a)

(ii) Some students needed more detail to "show" the equation of PQ.

(iii) Some students misquoted formula.

(iv) A number of very time-consuming solutions were provided for a quick 1 mark question - use of the diagram is a much quicker approach.

(v) Many students failed to connect their answers from (i) and (iii) using the area formula for a parallelogram $A=bh$.

(vi) There were a number of non-attempts - though this is not a difficult question. A diagram would be most useful. Correct rounding was required.

QUESTION THREE (continued)

$$(b) (i) \quad y' = \frac{(8-5x) \times 6x - (3x^2-4)x - 5}{(8-5x)^2}$$

$$= \frac{48x - 30x^2 + 15x^2 - 20}{(8-5x)^2}$$

$$\therefore y' = \frac{48x - 15x^2 - 20}{(8-5x)^2}$$

Comments for part (b)

(i) Most students applied the quotient rule correctly - though some solutions could have been tidier.

(ii) A variety of errors - mainly due to incomplete use of the product or chain rule.

$$(ii) \quad y = (10-3x)(1-4x^2)^{\frac{1}{2}}$$

$$y' = (10-3x) \times \frac{1}{2} (1-4x^2)^{-\frac{1}{2}} \times -8x + (1-4x^2)^{\frac{1}{2}} \times -3$$

$$= \frac{-4x(10-3x) - 3\sqrt{1-4x^2}}{\sqrt{1-4x^2}}$$

$$= \frac{-40x + 12x^2 - 3(1-4x^2)}{\sqrt{1-4x^2}}$$

$$\therefore y' = \frac{24x^2 - 40x - 3}{\sqrt{1-4x^2}}$$

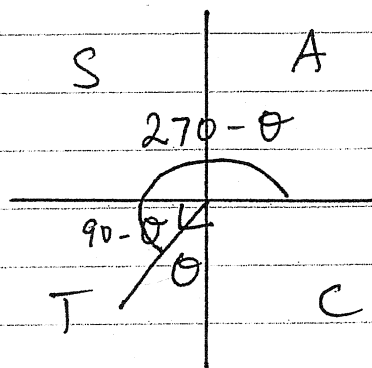
$$(c) \quad \text{LHS} = \frac{\sin(270-\theta)}{\sin\theta}$$

$$= \frac{-\sin(90-\theta)}{\sin\theta}$$

$$= \frac{-\cos\theta}{\sin\theta}$$

$$= -\cot\theta$$

$$= \text{RHS}$$



Comments for part (c)

Some students did not provide all steps to "show" the required result.

Communication of reasoning was a key element as to what was being examined in this question.

Question 4 - Year 11 - Mathematics - Task 3 2015

a) $y = x^2 - 8x + 20$ at $x = 2$, $y = 2^2 - 8(2) + 20$

$$\frac{dy}{dx} = 2x - 8 \quad \text{at } x = 2$$

$$y = 4 - 16 + 20$$
$$y = 8$$

$$\frac{dy}{dx} = 4 - 8$$

$$\therefore P(2, 8)$$

$$m = -4$$

$$\therefore \text{Eqn of tangent: } y - y_1 = m(x - x_1)$$

(This question
was completed
extremely well nearly
all students got it correct)

$$y - 8 = -4(x - 2)$$

$$y - 8 = -4x + 8$$

$$4x + y - 16 = 0$$

b) $\frac{\sin \theta}{1 - \sin \theta} - \frac{\sin \theta}{1 + \sin \theta} = 2 \tan^2 \theta$

$$\text{L.H.S} = \frac{\sin \theta}{1 - \sin \theta} - \frac{\sin \theta}{1 + \sin \theta}$$

$$= \frac{\sin \theta (1 + \sin \theta) - \sin \theta (1 - \sin \theta)}{(1 - \sin \theta)(1 + \sin \theta)}$$

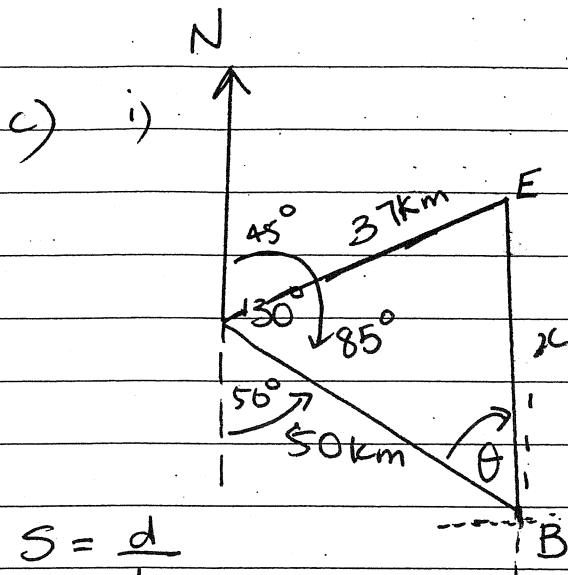
$$= \frac{\sin \theta + \sin^2 \theta - \sin \theta + \sin^2 \theta}{1 - \sin^2 \theta}$$

$$= \frac{2 \sin^2 \theta}{\cos^2 \theta}$$

$$= 2 \tan^2 \theta$$

$$= \text{R.H.S.}$$

(Most students
multiplied by the conjugate
of each fraction,
expanded & simplified
very well.)



$$S = \frac{d}{t}$$

$$S \times t = d$$

$$300 \times \frac{10}{60} = d$$

$$d = 50 \text{ km}$$

i) Diagram done well

ii) Most used

cosine rule some

substitute incorrect angle.

iii) Angle was found very well however, the bearing was poorly done.

$$ii) a^2 = b^2 + c^2 - 2bc \cos A$$

$$x^2 = 50^2 + 37^2 - 2(50)(37)\cos 85^\circ$$

$$x^2 = 3546.5$$

$$\therefore x = \sqrt{3546.5}$$

$$x = 59.55 \text{ km}$$

$$iii) \frac{a}{\sin A} = \frac{b}{\sin B}$$

$$\frac{37}{\sin \theta} = \frac{59.55}{\sin 85^\circ}$$

$$\frac{\sin \theta}{37} = \frac{\sin 85^\circ}{59.55}$$

$$\sin \theta = \frac{37 \times \sin 85^\circ}{59.55}$$

$$\sin \theta = 0.62$$

$$\theta = 38^\circ 14'$$

$$\therefore \text{Bearing} = N (50 - 38^\circ 14') W \text{ (alternate } \angle's)$$

$$= N 11^\circ 46' W$$

OR

$$348^\circ 14' T$$

$$OR/ \cos \theta = \frac{50^2 + (59.55)^2 - 37^2}{2 \times 50 \times 59.55}$$

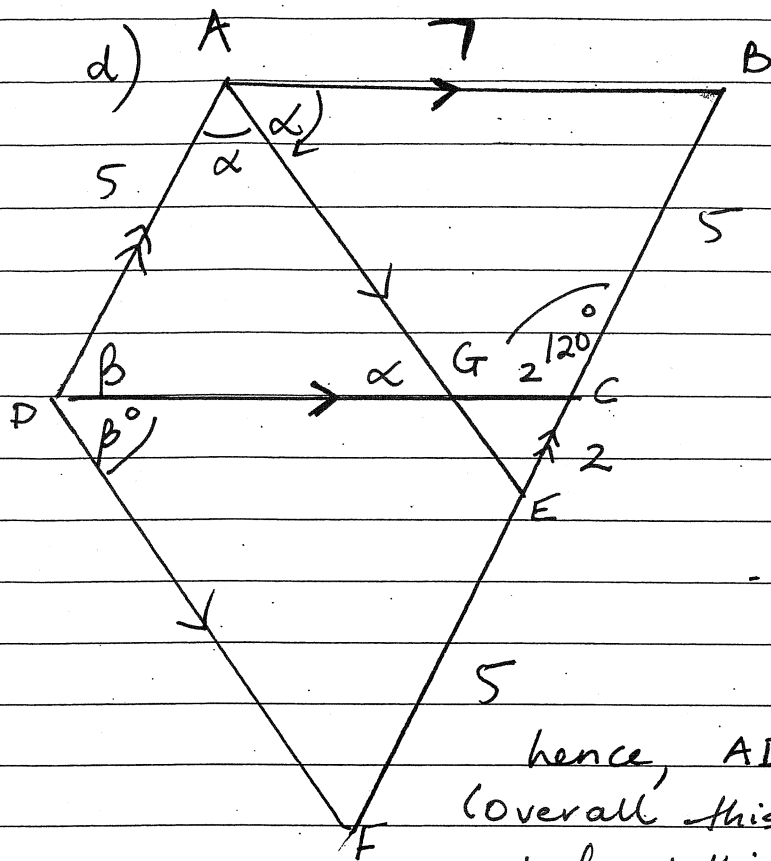
$$\cos \theta = 0.7854$$

$$\theta = \cos^{-1}(0.7854)$$

$$\theta = 38^\circ 14'$$

$$\therefore \text{Bearing} = N (50 - 38^\circ 14') W \text{ (alternate } \angle's)$$

$$= N 11^\circ 46' W \text{ or } 348^\circ 14' T$$



i) $\angle BAG = \alpha$ (given)
 $\angle AGD = \alpha$ (alt $\angle$'s
 $AB \parallel DC$)

$\angle ADC = \beta$ (given)
 $\angle FDG = \angle DGA$
 (alternate $\angle$'s, $DE \parallel AC$)

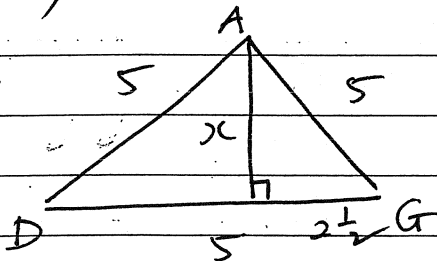
$$\therefore \alpha = \beta$$

$\therefore$ all angles in $\triangle ADG$
 are equal $\therefore \triangle ADG$
 is equilateral.

hence, $AD = AG = DG = 5$ units.

(Overall this question was done poorly
 most found this proof challenging)

ii) In $\triangle ADG$



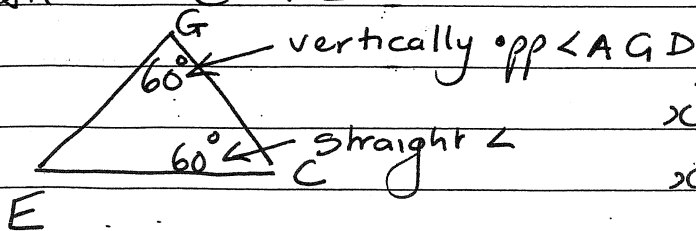
$$x^2 = 5^2 - 2\frac{1}{2}^2$$

$$x^2 = \frac{75}{4}$$

$$x = \frac{5\sqrt{3}}{2}$$

$DC = 7$ units opposite sides of parallelogram $ABCD$
 $\therefore GC = 7 - 5$
 $= 2$ units

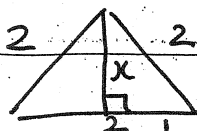
In $\triangle CGE$



$$x^2 = 2^2 - 1^2$$

$$x = \sqrt{3}$$

$\therefore \triangle CGE$ is equilateral



hence, Area of trapezium $= \frac{1}{2}h(a+b)$

$$= \frac{1}{2} \left(\frac{5\sqrt{3}}{2} + \sqrt{3} \right) (5+12)$$

$$= \left(\frac{5\sqrt{3}}{4} + \frac{\sqrt{3}}{2} \right) \times 17$$

$$= \left(\frac{5\sqrt{3} + 2\sqrt{3}}{4} \right) \times 17$$

$$= \frac{7\sqrt{3}}{4} \times 17$$

$$= \frac{119\sqrt{3}}{4}$$

Only a few students were successful in answering this part. Most who attempted didn't succeed. Most students ran out of time.